

Root Cause Graph — Analytical Methodology

Causal fault-propagation, alarm correlation, and root-cause ranking · NEXUS-1 analytics layer

Purpose of this note

The Root Cause Graph is the most method-bearing screen in the platform: unlike the read-once monitoring views, it makes explicit claims about how a cause is inferred. This note records the reasoning behind those claims — what the graph is, why it is built the way it is, and, equally important, what it deliberately does not do. It now covers three worked incidents, each demonstrating a distinct class of root-cause reasoning, and a section on how electromagnetic effects are (and are not) handled. It is a rationale document, not a UI walk-through; the in-application Help entry covers navigation.

The three-layer model

Causal structure in the graph is **engineered, not discovered from telemetry**. Inferring causal edges from observational time-series is unreliable under exactly the conditions a reactor presents — tight control-loop feedback, confounders, and sensor lag routinely manufacture spurious edges. The graph therefore rests on an auditable backbone, with data used only to weight and propose, never to define.

Layer	Source	Role
1 · Engineered backbone	Plant fault trees / event trees (PRA·PSA), component topology, and traceability records	Defines the causal and provenance edges. Every edge is traceable to an analysed failure path or a configuration record — auditable and regulator-aligned.
2 · Live runtime overlay	Active alarms and anomaly flags	Lights nodes by severity and animates propagation along the known edges. Adds state, not structure.
3 · Data-informed layer	Learned weights / candidate edges (ML)	Adjusts edge confidence and proposes candidate edges for engineer confirmation. Hints only — held out of the backbone until confirmed.

Four edge styles carry the meaning on screen: **solid** = engineered backbone (fault-tree or traceability); **dashed violet** = data-informed weight; **dashed orange** = artefact edge (an electrical/measurement disturbance corrupting a reading without a physical change); and **struck grey** = a candidate edge that was proposed and then rejected, or an expected corroboration that is absent. Node borders encode severity, not cause: **red** = origin or critical cascade, **amber** = degraded / transmitting, **green** = healthy or ruled out.

Three worked incidents — three reasoning classes

The screen carries three selectable incidents. Each is a hand-authored reconstruction chosen to exercise a different kind of inference, so that together they show the graph reasoning rather than merely drawing chains.

EVT-2026-0418 — forward propagation

A feedwater valve (FV-104) develops actuator latency. The graph traces the propagation to the protective trip four hops downstream:

FV-104 (latency) → SG-1 Level → FW Pump 2A (overspeed) → Pump 2A Bearing → Loop Flow (0.8 Hz) → Core ΔT → REACTOR TRIP

Reasoning class: **in-service degradation propagating forward** along a known chain, with an independent pump (RCP-1B) feeding the same loop-flow node as a parallel contributor. The terminal trigger is high core ΔT; the origin is the valve four hops upstream.

EVT-2026-0419 — real-vs-artefact discrimination (EMI)

A switchgear breaker (BKR-2A) closes and couples electrical noise into three instrument channels at the same instant. The readings look like a flow excursion, but the graph attributes the simultaneous, same-timestamp glitch to the switchgear transient (artefact edges) and flags that the independent instruments a real excursion would have moved — loop temperature and neutron flux — stayed flat. The struck-grey links to those steady instruments are the corroboration a genuine fault would produce, but which is absent here.

Reasoning class: **discrimination between a real process fault and a measurement artefact**. The tell is shared, simultaneous corruption across unrelated channels with no independent corroboration. The “real loss-of-flow” hypothesis is shown and ruled out.

EVT-2026-0420 — provenance / QA-escape tracing

A control rod (CR-7) that never met specification was admitted to service through an improperly dispositioned QA waiver, and surfaced months later as abnormal rod worth during a verification test. The graph follows the traceability chain back past the rod’s measured anomalies to the waiver itself — a records event that raised no control-room alarm. The two hypotheses an engineer would normally reach for first, drive-mechanism wear and a worth-prediction error, are shown ruled out: the mechanism checks healthy and the predictor is validated against the other banks.

Reasoning class: **tracing a latent fault back through provenance to a process origin**. The signature is that the anomaly was present from installation rather than developed in service, is isolated to one identified component, and the usual in-service explanations are excluded. The fault was admitted, not developed.

Two payoffs, common to all three

Alarm-flood suppression. Each incident collapses a burst of alarms into a single origin plus a grouped downstream cascade — 14→14%, 10→10% and 6 alarms reduced to one actionable origin, respectively. This is the ISA-18.2 alarm-management problem: the difference between an operator drowning and an operator oriented.

Causal-origin ranking beats alarm-count ranking. The strongest demonstration is a reversal that appears in every incident: the loudest node is rarely the cause.

Incident	Alarm-count ranks #1 (the trap)	Causal-origin ranks #1 (the cause)
0418	Pump 2A bearing — a downstream symptom	FV-104 valve latency — the upstream origin
0419	Flow transmitter FT-9 — the loudest channel	EMI artefact (BKR-2A) — a shared, uncorroborated disturbance
0420	Control rod CR-7 — the failing component	QA waiver WV-318 — a process escape, zero alarms

Confidences and rankings are illustrative, derived from edge weights, and are not identified from plant data.

Electromagnetic interference (EMI) and compatibility (EMC)

What they are. EMI is unwanted electrical noise coupling into a circuit; EMC is the property of equipment continuing to function correctly in the presence of that noise (and not emitting disruptive noise itself). A reactor’s instrumentation-and-control environment is electrically hostile: large motors, pumps, breakers, variable-frequency drives and high-current bus-work sit close to low-level instrument signals — thermocouples, neutron-flux detectors and transmitters reading millivolts. A switching transient can couple into those circuits and momentarily corrupt a reading or disturb logic.

Why it matters. Because some of those circuits are safety circuits, EMC is a qualification requirement, not a quality nicety. Digital instrumentation is more susceptible to interference than the analog and relay logic it replaced, so safety I&C; is EMC-qualified before service (in the United States under NRC Regulatory Guide 1.180, which endorses recognised MIL-STD and IEC emission and susceptibility test methods; IEC standards apply internationally), and resilience is built in through shielding, grounding, filtering, redundancy, voting logic and error-correcting memory.

A related phenomenon — single-event effects. High-energy particles from cosmic rays, and during solar storms from the Sun, can strike a microchip and flip a bit (a single-event upset). At ground level the atmosphere absorbs most of this flux so the rate is low, but it rises with altitude and during strong solar particle events, and across thousands of I&C; channels it is a real, if infrequent, consideration. Importantly, a radiation-induced upset produces the same diagnostic signature as EMI: a momentary, physically impossible reading that self-clears and is uncorroborated by independent instruments. It is therefore simply another origin for the same artefact-discrimination logic used in EVT-2026-0419 — no separate feature is required.

Scope boundary — what the platform does and does not do with EMC

EMC compliance is a **hardware-qualification responsibility** — it lives in cabling, shielding, grounding, filtering, enclosure design and the qualification of protection hardware, below the application. This analytics layer does not sense, test, or manage EMC, and makes no EMC compliance claim.

What the platform **does** do is represent EMI (and radiation-induced upsets) as a diagnostic fault class: an origin the root-cause graph can reason about, so an investigator can distinguish a measurement artefact from a genuine process fault. That is the whole of the EMC-related capability — a diagnostic aid, not a compliance function.

Real-world grounding

The three incidents are **representative of recognised industry concerns, not re-creations of specific accidents**. Forward-propagation feedwater/trip sequences are routine event-report material. Fuel and component handling/configuration errors are documented — a 2021 research-reactor event in which a fuel element was improperly seated and overheated is the closest public analogue to the configuration-control theme of EVT-2026-0420. The counterfeit-, fraudulent- and suspect-item (CFSI) premise behind that incident is a regulator-acknowledged concern (a 2022 NRC Inspector-General inquiry found such items present in operating plants), though regulators have also concluded no such item has been shown to have actually challenged reactor safety — a debated point. The scenarios are framed accordingly: they model real risk categories for demonstration, and should be described that way rather than as historical events.

Limitations & non-claims

- The causal structure is engineered from fault/event trees and traceability records — the platform does **not** perform causal discovery from telemetry, and learned edges never enter the backbone unconfirmed.
- Confidence scores are illustrative weightings for ranking and triage, not calibrated probabilities or a PRA result; the incidents are hand-authored reconstructions, not live detections.
- The graph is an **advisory and investigative** aid. It informs operators and investigators only; it has no control authority and never interacts with certified, hardware-isolated reactor protection systems.
- EMI / radiation upsets appear only as a diagnostic fault class; EMC qualification, and QA and procurement controls, are responsibilities outside this analytics layer.
- Output is a hypothesis set for engineer confirmation. A confirmed root cause feeds the incident record and the hash-sealed compliance log; the graph does not close an investigation on its own.